

WHITTIER, AK, USA

WMO: 702757

Lat: 60.777N

Lon: 148.719W

Elev: 39

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 94-05

WBAN: 26444

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.6	10.2	-5.8	4.0	9.9	-0.3	5.4	12.2	35.2	18.3	33.9	16.3	21.2	220	0.708

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	71.5	60.6	67.9	58.4	64.6	56.2	61.0	69.0	59.2	66.6	57.5	63.6	12.1	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
57.0	69.3	63.3	55.2	65.0	60.4	54.5	63.2	59.4	27.0	68.9	25.7	66.6	24.7	63.5	69.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.5	25.0	23.6	4.3	75.7	5.2	2.5	0.6	77.4	-2.5	78.9	-5.4	80.3	-9.2	82.1
			2.8	65.4	4.9	2.2	-0.7	67.0	-3.6	68.3	-6.3	69.5	-9.9	71.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.2	28.4	30.0	31.9	39.0	46.4	54.2	56.8	56.3	49.1	40.1	32.3	29.7
	DBStd	12.10	8.64	7.93	6.49	4.88	4.85	5.03	3.91	4.23	4.03	5.13	6.97	7.55
	HDD50	3801	671	559	562	333	133	13	0	1	61	308	530	630
	HDD65	8680	1136	979	1027	781	576	326	260	271	476	773	980	1095
	CDD50	603	0	0	0	2	22	138	210	196	35	0	0	0
	CDD65	8	0	0	0	0	0	2	5	1	0	0	0	0
	CDH74	16	0	0	0	0	0	4	10	2	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	8.5	10.5	9.5	9.8	8.3	7.3	6.5	5.6	6.6	8.1	9.3	10.2	10.3
	PrecAvg	161.7	14.5	13.7	12.2	13.0	11.6	7.7	9.2	12.0	18.3	17.3	16.5	16.9
Precipitation	PrecMax	195.3	39.0	26.0	23.4	27.9	27.5	13.2	16.7	24.3	39.8	36.1	42.7	31.9
	PrecMin	92.1	1.9	1.9	2.5	1.6	1.3	1.0	4.9	3.7	3.7	4.8	1.3	4.0
	PrecStd	23.0	8.3	6.7	6.1	6.3	7.7	3.0	3.2	5.7	8.5	7.1	11.1	7.0
	PrecStd	23.0	8.3	6.7	6.1	6.3	7.7	3.0	3.2	5.7	8.5	7.1	11.1	7.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	41.0	45.4	58.9	63.6	73.1	75.1	72.7	62.9	52.4	46.2	43.4
		MCWB	37.6	38.3	41.2	49.5	52.1	61.5	62.3	61.8	53.4	50.2	43.9	41.0
	2%	DB	41.1	39.1	42.6	53.9	61.2	70.2	71.7	69.9	59.7	50.0	44.8	41.5
		MCWB	37.9	37.6	38.9	46.3	50.4	59.1	60.6	60.3	51.8	47.2	42.7	39.7
	5%	DB	39.0	37.4	40.0	49.0	57.4	66.4	67.0	66.8	57.1	48.1	42.7	39.2
		MCWB	37.1	36.5	37.0	43.2	48.8	56.5	58.0	58.2	50.4	45.7	40.3	37.7
	10%	DB	37.2	36.9	38.3	45.8	55.0	63.7	63.9	64.2	55.0	46.3	39.8	37.4
		MCWB	36.0	36.0	36.0	40.7	47.8	55.0	56.5	56.2	49.9	44.2	37.9	36.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.3	39.4	42.3	50.6	54.2	63.1	64.0	63.8	55.0	51.7	45.7	42.6
		MCDB	41.7	40.0	44.4	58.5	61.1	72.4	72.9	71.0	59.3	52.2	45.8	43.7
	2%	WB	38.7	38.1	39.7	46.6	51.9	59.9	60.8	60.7	53.6	48.2	43.0	40.4
		MCDB	40.0	38.6	41.4	52.2	58.7	68.3	69.1	67.9	56.5	48.5	44.4	41.7
	5%	WB	37.4	37.2	37.9	43.8	50.3	57.4	59.0	59.1	52.5	46.4	40.7	38.4
		MCDB	38.4	37.6	39.2	48.5	56.7	65.2	66.4	65.7	55.2	46.7	42.3	39.3
	10%	WB	36.4	36.4	36.8	41.5	48.6	55.5	57.3	57.3	51.6	45.4	38.8	37.1
		MCDB	37.4	36.9	37.9	45.0	54.1	62.7	62.8	63.1	54.1	46.0	40.1	37.7
Mean Daily Temperature Range	5% DB	MDBR	6.9	6.3	7.8	9.3	10.9	12.0	9.9	10.2	8.0	6.6	6.5	7.2
		MCDBR	7.2	5.8	9.0	14.1	15.4	17.6	15.8	15.6	11.1	7.7	7.4	7.2
		MCWBR	6.2	5.3	7.4	10.8	10.5	11.6	10.0	10.0	7.4	6.5	6.4	6.7
	5% WB	MCDBR	6.3	5.4	8.2	12.9	13.9	15.8	13.8	14.7	8.3	6.4	7.0	6.7
		MCWBR	5.8	5.1	7.3	10.3	10.5	11.1	9.8	10.0	6.6	6.2	6.2	6.4
Clear-Sky Solar Irradiance	taub		0.249	0.312	0.326	0.369	0.380	0.383	0.388	0.373	0.323	0.311	0.255	0.240
	taud		2.054	2.063	2.074	2.017	2.115	2.243	2.306	2.346	2.474	2.266	2.040	1.914
	Ebn at Noon		152	203	246	257	266	267	263	257	255	207	142	92
	Edh at Noon		13	25	35	44	43	39	36	32	23	20	13	10
All-Sky Solar Radiation	RadAvg		96	315	815	1331	1660	1656	1492	1177	775	360	138	53
	RadStd		13	49	95	136	142	90	118	132	111	43	22	7

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (28 neighbors)	N/A	+3.42	+2.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page